EC24B1039 GAUTAM DEVARAJ <ec24b1039@iiitdm.ac.in>

Fwd: EEG Basics Assignment

1 message

Kannadasan K <kannadasankk@iiitdm.ac.in> Thu, Dec 4, 2025 at 12:19 PM
To: ec24b1039@iiitdm.ac.in

Dear Gowtham,
Finish the following assignment and send it to me within the next week. This will help you to understand the basics of signal processing.

You may refer to this for understanding basics: https://www.youtube.com/@mikexcohen1 (Go through the ones which are related to EEG)

Part 1: Generate a Simple Sine Wave

- Create a sine wave signal:
  - Duration: 5 seconds
  - Sampling Rate: 5 Hz
  - **Frequency: 5 Hz**
- Plot the signal in the time domain.
- Perform frequency analysis using the **Fourier Transform** (e.g., `np.fft.fft` in Python or `fft` in MATLAB).
- Plot the magnitude spectrum and interpret the dominant frequency.

Part 2: Generate Higher Frequency Sine Waves

Repeat the above steps for the following:

- Frequency: **10 Hz**
- Frequency: **20 Hz**

Note: You may increase the sampling rate (e.g., to **100 Hz**) for these signals to meet the Nyquist criterion and avoid aliasing.

Part 3: Combine Signals

- Combine the three sine waves (5 Hz, 10 Hz, 20 Hz) into a **single composite signal**.
- Plot the time-domain representation of the combined signal.
- Perform Fourier Transform and plot the frequency spectrum.
- Identify the presence of all three frequency components in the frequency domain.

Part 4: Add Noise

- Add **random noise** (e.g., Gaussian noise using `np.random.normal`) to the combined signal.
- Plot the noisy signal in the time domain.
- Perform Fourier Transform and plot the frequency spectrum.
- Discuss how noise affects the frequency representation.

Part 5: Filtering

- Apply a **bandpass filter** (e.g., 0.5–30 Hz) to the noisy signal and analyze the result.

- Compare the spectrum of the raw noisy signal and the filtered signal.

Submission Requirements:

- Python script or Jupyter Notebook / MATLAB Live script (with all details)
- Time-domain and frequency-domain plots for each task
- Short interpretation/summary of each frequency spectrum

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Regards,
Dr. Kannadasan K Ph.D.,